

Integral Relations among Totient Sections

Finite-level bases and bounded residue series

WILL COOK*

July 2026; revised 29 September 2026

ABSTRACT

We give integral coordinates for the sections $n \mapsto \varphi(k^j n + r)$ through any fixed level $e \geq 1$, in every integer base $k \geq 2$. A retained family of $k^e + 1$ sections is a basis, and the scalar reductions of the other indexed sections form a basis of all integral relations. We give a finite-evaluation proof of independence and explain its relation to Martin's stronger affine theorem. Prime isolation also classifies rational binary series of dyadic totient residues. The irrationality of the unreduced binary totient series remains open.

1 Introduction

For a fixed integer base $k \geq 2$, consider the sequences $n \mapsto \varphi(k^j n + r)$ with $j \geq 0$ and $0 \leq r < k^j$, called the *totient sections*. We take $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$. The identities

$$\varphi(4n) = 2\varphi(2n), \quad \varphi(4n + 2) = \varphi(2n + 1)$$

reduce the seven sections through level two in base two to five. We determine every relation in this way: reduce each section to a retained section, then compare its coefficient in that family. When listing the relations we keep the original indices distinct; for example, the equal sequences $\varphi(4n + 2)$ and $\varphi(2n + 1)$ still give two generators.

The sections form the k -kernel of the totient sequence. A sequence is k -regular when the $\mathbb{Z}$ -module generated by its k -kernel is finitely generated [1]. Coons proved that the totient sequence is not k -regular for any $k \geq 2$ [2, Thm. 3.2]. The finite-level description below gives the individual reductions and the resulting integral coordinates. At level zero the sole section is $\varphi(n)$, of dimension one.

Theorem 1.1 (A basis through each finite level). [Lean](#) · [Comparator](#) *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

[Lean](#)
[Comparator](#)

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{F_{0,0}^{(k)}, F_{1,0}^{(k)}\} \cup \{F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r\}$$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1}F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

The condition $k \nmid r$ allows residues with $\gcd(k, r) > 1$. For instance, $r = 1$ and $r = 2$ are both retained in base six, and the reductions involving them are

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The second multiplier includes the factor 2 already present in $6n + 2$. Repeated prime factors of k cause no further exception to the product formula, so the theorem also applies to nonsquarefree bases.

Martin's 2006 theorem on simultaneous totient inequalities already implies independence for every finite family of pairwise nonproportional affine forms with positive integral slopes [4, Thm. 1]. Its conclusion is stronger than the independence used here. The point of the present description is the explicit reduction of the sections and the integral relation basis in Corollary 3.1. We also give a direct proof that constructs a nonsingular evaluation matrix. It uses the CRT–Dirichlet separation argument in Yazdani's proof of Theorem 2, credited there to Shallit [12, pp. 652–653]. The step that needs care is to keep the progression for the selected affine form reduced while imposing prime divisors on all the others.

Section 2 proves the basis theorem. Section 3 starts with the six reductions in base six and derives the integral relation basis. In Section 4 we use prime isolation again, for bounded functions of totient residues. The series

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

has unbounded coefficients. Its irrationality, asked in Erdős Problem #249 [10, p. 61], remains open. The coefficient basis supplies no irrationality conclusion for this value. Appendix A gives further coefficient consequences; Appendices B and C collect the tail identities, comparison sequences and still-unproved cancellation estimates relevant to S . Their longer proofs are in *The Binary Totient Series*.

2 Proof of the basis theorem

To prove spanning, we use the product formula for φ to keep track of the primes introduced by multiplication. When $n > 0$, the integers kn and $k^j n$ have the same prime divisors, giving $\varphi(k^j n) = k^{j-1} \varphi(kn)$, and at $n = 0$ both sides vanish. For $r = k^t u > 0$ with $k \nmid u$, a prime $p \mid k$ divides $k^{j-t} n + u$ exactly when it divides u . Multiplication by k^t therefore contributes the factor k^t and the additional factors $1 - 1/p$ for $p \mid k$, $p \nmid u$, giving $C_k(t, u)$.

For independence, Martin's theorem is enough. For any ordering of pairwise nonproportional affine forms and any $C > 0$, it gives a set of positive lower density on which

each successive totient ratio exceeds C . In a proposed nonzero relation, place a nonzero coefficient c_1 first and take $C > \max(1, \sum_{i>1} |c_i|/|c_1|)$. The first term then dominates the sum of all remaining terms in absolute value, a contradiction [4, Thm. 1]. This argument also proves independence after any finite prefix is deleted.

We give a second proof because it produces a square evaluation matrix whose determinant is nonzero modulo one prime. Consider affine forms $L_i(n) = a_i n + b_i$ ($1 \leq i \leq s$) with $a_i > 0$ and $a_i b_j - a_j b_i \neq 0$ for $i \neq j$. A shift of the variable makes every b_i positive without changing the cross determinants, after which we write $g_i = \gcd(a_i, b_i)$ and $A_i = L_i/g_i$. We seek evaluation points at which just one of these totients is nonzero modulo a fixed prime ℓ . To obtain the point for column i , we will make $A_i(n)$ prime while giving every other form a prime divisor congruent to 1 modulo ℓ .

We choose an odd prime ℓ dividing none of the a_i , the $g_i \varphi(g_i)$ or the nonzero cross determinants. For each i we then choose distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ ($j \neq i$) outside the same finite exceptional set, using Dirichlet's theorem. The congruences

$$A_i(n) \equiv 2 \pmod{\ell}, \quad L_j(n) \equiv 0 \pmod{q_{ij}} \quad (j \neq i)$$

are individually soluble and have pairwise coprime moduli, so CRT combines them into a progression $n = n_0 + Ru$, where $R = \ell \prod_{j \neq i} q_{ij}$.

The progression for $A_i(n)$ has modulus $(a_i/g_i)R$. We check that it is reduced before applying Dirichlet's theorem. Primitivity excludes every prime factor of a_i/g_i , and $A_i(n) \equiv 2 \pmod{\ell}$ excludes ℓ . Finally, if $q_{ij} \mid A_i(n)$, then q_{ij} divides both $L_i(n)$ and $L_j(n)$, and consequently divides $a_i b_j - a_j b_i$, which was excluded when q_{ij} was chosen. These are all the prime factors of the modulus, so Dirichlet's theorem supplies arbitrarily large prime values $p = A_i(n)$. We choose one with $p \nmid g_i$.

At the resulting point n_i ,

$$\varphi(L_i(n_i)) = \varphi(g_i)(p-1) \equiv \varphi(g_i) \not\equiv 0 \pmod{\ell}.$$

For $j \neq i$ we have $q_{ij} - 1 \mid \varphi(L_j(n_i))$, so every other entry of row i vanishes modulo ℓ . We may choose the points n_i independently and beyond any prescribed threshold; the resulting evaluation matrix is diagonal modulo ℓ , with nonzero diagonal entries, and therefore has a nonzero integer determinant. This proves rational independence of the sequences $\varphi(L_i(n))$ after deletion of any finite prefix. In particular, the argument requires only one prime value in each row.

It remains to separate $\varphi(n)$ and $\varphi(kn)$, whose affine arguments are proportional. On the progression $n = km + 1$ we have $\varphi(kn) = \varphi(k)\varphi(n)$, so a proposed relation contains only one zero-residue term after this restriction. The positive-residue sections become the forms

$$k^{j+1}m + k^j + r \quad (k \nmid r),$$

together with $km + 1$. These are pairwise nonproportional: equality of two intercept-to-slope ratios would give $r/k^j = r'/k'^j$, which for $j' > j$ forces $k \mid r'$. No positive-residue form is proportional to $km + 1$. Affine independence eliminates every positive-residue coefficient and leaves $A + B\varphi(k) = 0$, where A, B multiply $\varphi(n), \varphi(kn)$. We now evaluate the remaining relation at $n = k$ to obtain $A + Bk = 0$. Since $\varphi(k) < k$, the two equations

force $A = B = 0$. Finally,

$$|\mathcal{B}_{k,e}| = 2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1,$$

which completes the proof of Theorem 1.1. For independence on a tail, use a sufficiently large power of k in the last evaluation.

3 Integral relations

The scalar reductions make an equality test possible without choosing evaluation points. We first carry it out in base six at level two, where there are 43 indexed sections. The 37 retained sections are $F_{0,0}, F_{1,0}$, the five $F_{1,r}$ ($1 \leq r \leq 5$), and the 30 sections $F_{2,r}$ with $1 \leq r < 36$ and $6 \nmid r$. The omitted sections satisfy

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

Thus an integral combination of the 43 sections vanishes precisely when these six substitutions leave all 37 retained coefficients zero. The six differences obtained by moving the right-hand sides to the left give every integral relation. Indeed, subtracting the original coefficient of each omitted section times its difference removes all six omitted coordinates. The remaining retained coordinates must then vanish by Theorem 1.1. This is the general argument of the next corollary. At base two and level two the corresponding relation basis is simply $E_{2,0} - 2E_{1,0}$, $E_{2,2} - E_{1,1}$.

Corollary 3.1 (Integral coordinates and all integral relations). [Lean · Comparator](#) *Let $k \geq 2$ and $e \geq 1$. The retained family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Index the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index i , write the scalar reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group with one generator E_i for each of these indices, the vectors*

[Lean](#)
[Comparator](#)

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. Since $\prod_{p|k, p \nmid u} p$ divides k , every scalar $C_k(t, u)$ is an integer. The retained family therefore spans the section module over $\mathbb{Z}$, and its rational independence makes it a $\mathbb{Z}$ -basis.

Given $x = \sum_i x_i E_i$ in the kernel, we subtract $\sum_{i \text{ omitted}} x_i R_i$ to cancel all its omitted coordinates. The resulting vector is supported on the retained indices and still evaluates to zero, hence vanishes by independence. Conversely, the coefficient at omitted index i in $\sum_h c_h R_h$ is c_i itself, so we recover every coefficient without division. This proves both independence and uniqueness. Counting omitted indices gives $(1+k+\dots+k^e) - (k^e+1) = \sum_{j=1}^{e-1} k^j$. $\square$

The assertion concerns the module generated by the sections. It need not contain every integer-valued sequence in its rational span: for $k = 2$ and $e \geq 2$, the sequence $\varphi(4n + 3)/2$ is integer-valued and has coefficient $1/2$ in the retained basis. Thus this module is not saturated in the group of integer-valued sequences.

We can also decide equality by finitely many evaluations. Order the retained columns as above, with residues increasing within each level, and evaluate at $n = 0, \dots, 36$ to obtain a 37×37 matrix. Its determinant is 17 modulo 101, so it is non-singular over $\mathbb{Q}$ and detects every nonzero retained combination at one of these inputs. The script `scripts/totient_kernel_normal_form.py` calculates the coordinates and the first such input using integer arithmetic. The equality test uses the integer evaluations. Reducing the proposed combination modulo 101 would miss $101F_{2,2}$, for example, although its value at $n = 0$ is 101.

4 Bounded functions of totient residues

We now use the same congruence construction to separate an infinite series from the rationals. Reducing $\varphi(n)$ modulo a fixed integer makes its coefficients bounded, which allows a long block of zero residues to control the entire remaining tail. Residues in the following sum are their least nonnegative representatives.

Theorem 4.1 (Residue series and dyadic observables). [Lean](#) · [Comparator](#) *For every $m \geq 3$,*

[Lean](#)
[Comparator](#)

$$\sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k)2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Modulo two the sum is $3/4$, since $\varphi(n)$ is even for $n \geq 3$. More generally only $f(1)$ and the even residue classes occur in the second assertion. Its proof uses the following elementary tail argument.

Lemma 4.2 (An isolated nonzero coefficient between long zero blocks). [Lean](#) · [Comparator](#) *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

[Lean](#)
[Comparator](#)

Proof. Suppose that the sum has denominator $q \geq 1$, so that every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1}\mathbb{Z}$. We choose an isolated coefficient with $C2^{-L} < 1/q$. The zero block to its right gives $|T_N| \leq C2^{-L} < 1/q$ and therefore $T_N = 0$, whereas the block to its left then gives $T_{N-L-1} = a_N 2^{-L-1}$, a nonzero element of $q^{-1}\mathbb{Z}$ of absolute value less than $1/q$. This is a contradiction. $\square$

For example, the indicator of the squares satisfies the lemma. An eventually periodic sequence with infinitely many nonzero terms does not have isolated coefficients at arbitrarily large distances. The uniform bound C is essential to this argument.

Proof of Theorem 4.1. Fix a unit $s \pmod{m}$ and a positive integer L . Choose distinct primes $q_t \equiv 1 \pmod{m}$, all exceeding L , for $0 < |t| \leq L$. The congruences

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t} \quad (0 < |t| \leq L)$$

define a reduced progression, since s is a unit modulo m and $0 < |t| < q_t$. CRT and Dirichlet's theorem therefore allow us to choose an arbitrarily large prime $p > L + 1$ on this progression, with $m \mid \varphi(p + t)$ at every neighbour and $\varphi(p) \equiv s - 1 \pmod{m}$ at the centre.

Subtract $f(0)$ and multiply by a common denominator of the values of f . If $f(s - 1) \neq f(0)$, the resulting bounded integer sequence has an isolated nonzero coefficient for every L , so the lemma applies. For the least-residue map we take $s = -1$ and obtain the nonzero residue $m - 2$ when $m \geq 3$. When $m = 2^k$, each even residue r is of the form $s - 1$ with s a unit, so nonconstancy on the even classes gives an irrational sum by the same argument. Conversely, if f is constant there, all terms after $n = 2$ equal c and we obtain $3f(1)/4 + c/4$. $\square$

The first assertion is also an application of Erdős's sparse-series criterion [6, Lem. 1, p. 213], and of Kaneko, Suzuki and Tachiya [7, Thm. 3]. The nonzero residues occur on a set of density zero: an integer in that set has no prime divisor $p \equiv 1 \pmod{m}$. Wong gave the case in which the base equals the modulus [5]. The isolated-coefficient argument above also treats signed values of $f - f(0)$, for which those nonnegative sparse-series criteria need an additional spacing hypothesis. The [extension to other bases](#) treats every integer $t \geq 2$ and proves rational linear independence of 1 and $\sum_n (\varphi(n) \bmod m) t^{-dn}$, $d \geq 1$.

A Further consequences for the sections

The finite-level theorem also describes the full dyadic kernel. The notation below specialises the base to two.

Corollary A.1 (Dyadic basis). [Lean](#) · [Comparator](#) *The family*

[Lean](#)
[Comparator](#)

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Each finite subset of this family is contained in some $\mathcal{B}_{2,e}$. Theorem 1.1 gives independence, and its reductions give spanning. At level zero the sole section $\varphi(n)$ is nonzero, since $\varphi(1) = 1$. $\square$

The affine argument also permits periodic coefficients by restriction to their common residue classes.

Corollary A.2 (Periodic coefficients). [Lean](#) · [Comparator](#) *Let $L_1, \dots, L_s$ be pairwise nonproportional affine forms with integer coefficients and positive slopes. If $w_1, \dots, w_s$ are rational-valued periodic sequences, then*

[Lean](#)
[Comparator](#)

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \implies w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. We choose a common period Q and restrict to $n = r + Qt$. The coefficients are now the constants $w_i(r)$, and the cross determinants are $Q(a_i b_j - a_j b_i) \neq 0$. By shifting t beyond the given threshold we can apply affine independence, obtaining $w_i(r) = 0$ for every i . We repeat this for each residue r ; periodicity then gives the conclusion even at the indices preceding the threshold. $\square$

Proportional forms are excluded for a reason: the identity $\varphi(4n + 2) = \varphi(2n + 1)$ would otherwise contradict the conclusion. Martin's theorem gives the same corollary on each progression [4, Thm. 1].

Coons's proof of nonregularity counts the poles of $\zeta(s - 1)/\zeta(s)$ [2, Cor. 3.1, Thm. 3.2]. For an integer-valued sequence, finite dimension of the rational kernel span is equivalent to finite generation of its integral kernel module: choose finitely many evaluations that embed the span in $\mathbb{Q}^d$; the module embeds in $\mathbb{Z}^d$, whose subgroups are finitely generated (compare [3, App. A]). Theorem 1.1 recovers this nonregularity and gives the ranks of its finite truncations. For an account of the earlier separation method, see Allouche, Shallit and Yassawi [13, Thm. 3, Ex. 4, Rem. 7].

These facts have a functional consequence. A k -Mahler equation has the form $\sum_{i=0}^d P_i(z)F(z^{k^i}) = 0$, with $P_i \in \mathbb{Q}[z]$ and $P_0 \neq 0$. The height theorem of Adamczewski, Bell and Smertnig implies that a k -Mahler series with integer coefficients of polynomial growth is k -regular [16, Thm. 1.2(b)]. Since $\varphi(n) \leq n$, its logarithmic height is at most $\log n$, so $F_\varphi(z) = \sum_{n \geq 1} \varphi(n)z^n$ has no such equation. Bell and Smertnig's multiplicative classification gives the same conclusion [14, Thm. 1.3]. Bell, Bruin and Coons treat related algebraic and D -finite series, the latter satisfying a linear differential equation with polynomial coefficients [15, Thms. 1.5–1.6]. These statements concern the function, and do not determine $F_\varphi(1/2)$.

The section operators $T_i g(n) = g(kn + i)$ satisfy

$$T_i F_{j,r} = F_{j+1,ki+r}, \quad V_{k,e+1} = V_{k,e} + \sum_{i=0}^{k-1} T_i V_{k,e}.$$

To obtain the second identity, write every residue at level $j + 1$ as $ki + r$. For a positive retained residue, the resulting section is again retained. For the two zero-residue generators, use $T_i F_{0,0} = F_{1,i}$ and $T_i F_{1,0} = F_{2,ki}$ followed by the zero-residue or scalar reduction. We thus obtain a positive integer multiple of a basis element in every case, although the increasing dimensions prevent any finite truncation from being invariant. The growth formula using the joint spectral radius for finite-dimensional regular sequences [3, Thm. 1, Prop. 4, Cor. 7] therefore does not apply to φ .

B The binary totient series

The proof of Theorem 4.1 uses boundedness where a long zero block makes the tail small. For S , the estimate $0 \leq \varphi(n) \leq n$ proves convergence, but forcing $2^L \mid \varphi(N + L)$ with $N + L > 1$ already requires $2^L \leq N + L - 1$. The tail bound $(N + L + 2)2^{-L}$ then exceeds 1. Thus this divisibility construction does not supply the small nonzero tail needed to exclude a rational value. We consider instead a finite test for differences of scaled tails.

Put $R_N = \sum_{j \geq 1} \varphi(N+j)2^{-j}$ and $\Delta_h(N) = R_{N+h} - R_N$. Subtracting the finite-prefix identities for $2^{N+h}S$ and $2^N S$ gives

$$\Delta_h(N) \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}. \quad (1)$$

If $S = a/(2^c v)$, with v odd, then $\Delta_h(N)$ is integral for $N \geq c$ and $h = \varphi(v)$. To disprove rationality it therefore suffices to find a nonintegral difference beyond every threshold for each positive h .

For $h, N, L \in \mathbb{N}$, define the integer

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j))2^{L-1-j}, \quad B = N+h+L+2.$$

Since $0 \leq R_M \leq M+2$, we have

$$2^L \Delta_h(N) - D_{h,N,L} = R_{N+h+L} - R_{N+L}, \quad |2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (2)$$

The difference is bounded by the larger tail bound. Hence the finite inequality

$$B < D_{h,N,L} \pmod{2^L} < 2^L - B \quad (3)$$

forces $\Delta_h(N) \notin \mathbb{Z}$. Conversely, if its distance to $\mathbb{Z}$ is positive, choose L so that $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$; the same error estimate gives (3). Thus allowing an unrestricted depth loses no nonintegral difference. For example, $D_{1,12,16} = -143140$ has residue 53468 modulo 2^{16} , and $B = 31$, so it excludes denominators dividing 2^{12} .

B.1 Equivalent residue conditions

We write $K(h, N, L)$ for (3).

Theorem B.1 (Propagation of one nonintegral tail difference). [Lean · Comparator](#) Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently

[Lean](#)
[Comparator](#)

$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

The proof uses the recurrence $\Delta_{(t+1)d}(N) \equiv 2^d \Delta_{td}(N) + \Delta_d(N) \pmod{\mathbb{Z}}$; it is given in [Appendix A of the long record](#). A single pair d, N need not imply irrationality; the final assertion requires every positive d and every $N \geq 0$.

For a hypothetical denominator $2^c v$, a second test uses the modulus $M = (2^H - 1)/v$, where H is a positive multiple of $\varphi(v)$. Write $B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j)2^{H-1-j}$.

Equivalent formulation B.2 (A residue test modulo a divisor of $2^H - 1$). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$c + H + 1 < (-B_{H,c}) \pmod{M} < M - (c + H + 1). \quad (4)$$

The [proof and residue recurrence](#) use $B_{H,c} = M(vR_c) - \Delta_H(c)$ and the linear tail bound. Neither test provides these universal witnesses. A separate [Farey-interval calculation](#) at depth 240 excludes $S = a/q$ for $0 < q \leq Q_0 \approx 7.96 \times 10^{34}$.

B.2 Comparison sequences

Large ranks alone cannot settle the series question. There is an integer sequence c with $0 \leq c(n) \leq n$, unchanged odd totients and $|c(n) - \varphi(n)| \leq 2$ at even indices, for which $\sum c(n)2^{-n} = 5/4$. Its integral carry still has dyadic rank at least $2^e - 1$ through every positive level. The [construction and its limitations](#) explain why it fails the exact prime-dilation laws of φ . The same section retains the cofinal LCM-diagonal criterion, its endpoint sign conditions and the limits of finite integer combinations of shifts.

The ordinary [signed interpolation argument](#) permits any value in an interval about $\sum \varphi(n)\beta^{-n}$, for each integer $\beta \geq 2$, while preserving a prescribed prefix and every odd coefficient, with errors bounded by any positive $f(n) \rightarrow \infty$. The changed even indices have upper Banach density zero. On each fixed progression the Dirichlet defect is entire; at roots of unity the generating-function defect has smooth radial data and any prescribed finite jets vanish. This argument has not been formalised in Lean. It preserves Dirichlet singularities; regular parts and zeros may change, and finite jets do not determine an entire radial germ. It therefore supplies no cancellation estimate for the unchanged totient.

Rational approximation imposes a different requirement. The identity

$$\Theta_2 := \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}$$

was posted by Amiram Eldar, together with its coprimality interpretation, on 15 March 2026 [8], and by Steve Fan on 16 May 2026 [9]. Consider the rational quotients

$$Q(e, Y) = \frac{(\sum_{d \leq Y} \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d \leq Y} \mu(d)/(2^d - 1)^{2e+2}}.$$

Theorem B.3 (A lower bound for the rank-one quotients). [Lean · Comparator](#) For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies

[Lean](#)
[Comparator](#)

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

The [proof](#) bounds geometric tails and compares seven-term rational prefixes. The averaging assertion applies after division; signed combinations, coupling before division and higher-rank kernels remain separate questions. A proposed family p_j/q_j in lowest terms would need $0 < q_j|\Theta_2 - p_j/q_j| \rightarrow 0$, which convergence alone does not imply.

C An exponential-sum question

For every $h \geq 1$ and every X_0 , suppose that some integers $X \geq \max(X_0, 1)$ and $L \geq h$ satisfy

$$16(2X + h + L + 2) \leq 2^L, \quad \operatorname{Re} \sum_{N=X}^{2X-1} \exp(2\pi i D_{h,N,L}/2^L) \leq \frac{9}{10}X. \quad (5)$$

These inequalities would prove irrationality. Indeed, if every residue failed (3), each phase would be within $1/16$ of an integer, so every real part would be at least $\cos(\pi/8) > 9/10$. A certificate must therefore occur on the block, and (1) excludes every rational denominator by choosing its period and then a sufficiently late block.

To separate the counting estimates from the cancellation estimate, we single out one totient in each window. Fix $h, s \geq 1, L \geq s + h, X \geq 1$ and $0 < \eta < 1$, and put $t = L - s + 1$ and $E_N = \exp(2\pi i D_{h,N,L}/2^L)$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and $m_N = (N + t)/p_N$. Define

$$\begin{aligned} \mathcal{A} &= \{N \in [X, 2X) : m_N \leq \lfloor \sqrt{X}/2 \rfloor, p_N > 2\lfloor \sqrt{X} \rfloor\}, \\ \mathcal{G} &= \{N \in \mathcal{A} : \varphi(m_N) \geq \eta m_N\}. \end{aligned}$$

All indices are integers. On $\mathcal{A}$ we have $p_N > m_N$, so $\varphi(N + t) = \varphi(m_N)(p_N - 1)$. Its coefficient in the phase is $(2^h - 1)/2^t$. Write

$$z_N = \exp\left(\frac{2\pi i (2^h - 1) \varphi(m_N)(p_N - 1)}{2^t}\right), \quad w_N = E_N / z_N,$$

and let $\bar{z}_m$ be the average over the indices in $\mathcal{A}$ with cofactor m , with value zero for an empty set. The four proposed bounds are

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25} X, \\ \left| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right| &\leq \frac{1}{100} X, \\ \left| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E_N \right| &\leq \frac{1}{100} X, \\ \left| \sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E_N \right| &\leq \frac{8}{25} X. \end{aligned} \tag{6}$$

Before taking real parts and absolute values, the four sums add to $\sum E_N$, and hence their bounds imply the real-part estimate $9X/10$ in (5). We have only added and subtracted the group means, without any independence assumption. The resulting bound concerns the real part, and does not give a $21X/25$ bound for the absolute value of the full sum.

Problem C.1 (Four estimates for the first-harmonic sum). For each $h \geq 1$, do there exist $s \geq 1$ and $0 < \eta < 1$ such that, for every X_0 , there are X, L with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

All four estimates must hold on the same block with the same parameters. For each fixed h, s , take the least admissible L satisfying the room inequality and $L \geq s + h$. The [smooth-number and small-cofactor estimates](#) give the last two bounds for all sufficiently large X when $0 < \eta < 1/(300 \log 4)$. The elementary excluded-cofactor estimate is $(3\eta \log 4 + o(1))X$; it needs no prime number theorem. The group-mean and weighted mean-subtracted estimates remain open. No optimality is asserted for the sufficient margins in (6).

Alternatively, at $s = 26$, $\eta = 1/1000$ and minimal depth one has $|\mathcal{G}| > 67X/100$ for large X . The single open estimate $\operatorname{Re} \sum_{N \in \mathcal{G}} E_N \leq 603X/1000$, for each h on arbitrarily large blocks, would suffice by the same argument. The [ordinary comparison argument](#) gives larger first harmonics at all sufficiently large admissible scales, distinguishing its rational sequences from the cancellation required of φ . Its proof is included with the interpolation construction.

Balasubramanian, Giri and Srivastav give correlation estimates applicable to $\varphi(n)/n$, uniformly for $|h| \leq x/2$ [[11](#), Thm. 2.2]. Here the modulus grows with L , and the residual weight depends on other totient values in the window. That correlation theorem supplies neither of the two remaining estimates. Establishing them for the actual totient is the unresolved arithmetic step.

Verification. *Formal proofs.* A result with a kernel-checked Lean proof carries links beside its heading and in the margin; both open the same evidence. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. Inline citations to individual declarations elsewhere may identify ingredients of a proof. *Comparator* opens the recorded comparison, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. *Comparator pending* means that the complete statement has no recorded comparison yet. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. Comparator checks statements, axioms and kernel acceptance; it does not establish novelty or peer review. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text. The [declaration guide](#) preserves the individual formal-source links for the short paper, including the three results whose full proofs appear in Appendix A of the long record. The displayed arguments and the signed interpolation proof are to be read as ordinary mathematical proofs. Recorded Lean and Comparator checks refer to their exact declarations and source versions; this revision makes no claim of a new formal run or independent review.

Acknowledgement. I thank Wouter van Doorn for advice on writing for a first-time reader and on explaining the strength of a hypothesis. His advice concerned a different note; it was not a mathematical review or endorsement of the results presented here.

References

- [1] J.-P. Allouche and J. Shallit, *The ring of k -regular sequences*, Theoret. Comput. Sci. **98** (1992), no. 2, 163–197, doi:[10.1016/0304-3975\(92\)90001-V](https://doi.org/10.1016/0304-3975(92)90001-V).
- [2] M. Coons, *(Non)Automaticity of number theoretic functions*, J. Théor. Nombres Bordeaux **22** (2010), no. 2, 339–352, doi:[10.5802/jtnb.718](https://doi.org/10.5802/jtnb.718). Theorem 3.2, p. 348.
- [3] M. Coons, *Regular sequences and the joint spectral radius*, Internat. J. Found. Comput. Sci. **28** (2017), no. 2, 135–140, doi:[10.1142/S0129054117500095](https://doi.org/10.1142/S0129054117500095); arXiv:[1511.07535v1](https://arxiv.org/abs/1511.07535v1). Theorem 1, Proposition 4, Corollary 7 and Appendix A are cited with the arXiv v1 numbering.
- [4] G. Martin, *Simultaneous inequalities among values of the Euler phi-function*, arXiv:[math/0603053v1](https://arxiv.org/abs/math/0603053v1), 2006. Theorem 1, pp. 1–2.
- [5] Erick Wong, [answer 1211557](https://math.stackexchange.com/a/1211557) to *An infinite sum based on the mod-parity of Euler’s totient function*, Mathematics Stack Exchange, 29 March 2015, accessed 15 September 2026.
- [6] P. Erdős, *On the irrationality of certain series*, Indag. Math. **19** (1957), 212–219. Lemmas 1 and 4, pp. 213–218.
- [7] H. Kaneko, Y. Suzuki and Y. Tachiya, *Refinements of Erdős’s irrationality criterion for certain sparse infinite series*, arXiv:[2601.20743v1](https://arxiv.org/abs/2601.20743v1). Theorem 3, p. 5.
- [8] A. Eldar, comment and formula added to OEIS A256936 (revisions 28 and 31), 15 March 2026, <https://oeis.org/history?seq=A256936>, accessed 16 September 2026.
- [9] S. Fan, comment on Erdős Problem #249, 16 May 2026, 19:01, [Erdős Problems discussion thread](https://mathoverflow.net/questions/478249/erdos-problem-249), accessed 16 September 2026.
- [10] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L’Enseignement Mathématique **28**, 1980, p. 61.
- [11] R. Balasubramanian, S. Giri and P. Srivastav, *On correlations of certain multiplicative functions*, J. Number Theory **174** (2017), 221–238, doi:[10.1016/j.jnt.2016.10.001](https://doi.org/10.1016/j.jnt.2016.10.001); arXiv:[1511.02221v3](https://arxiv.org/abs/1511.02221v3). Theorem 2.2 is cited with the arXiv v3 pagination, p. 2.
- [12] S. Yazdani, *Multiplicative functions and k -automatic sequences*, J. Théor. Nombres Bordeaux **13** (2001), no. 2, 651–658, https://www.numdam.org/item/JTNB_2001__13_2_651_0/. Theorem 2 and its proof, pp. 652–653; Corollary 4, p. 654.
- [13] J.-P. Allouche, J. Shallit and R. Yassawi, *How to prove that a sequence is not automatic*, Exposition. Math. **40** (2022), no. 1, 1–22, doi:[10.1016/j.exmath.2021.08.001](https://doi.org/10.1016/j.exmath.2021.08.001); arXiv:[2104.13072v1](https://arxiv.org/abs/2104.13072v1). Theorem and example numbering follows the arXiv v1 (2021).
- [14] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, arXiv:[2603.23456v1](https://arxiv.org/abs/2603.23456v1), 24 March 2026, preprint. Theorem 1.3, p. 2; Lemma 3.3, p. 9.
- [15] J. P. Bell, N. Bruin and M. Coons, *Transcendence of generating functions whose coefficients are multiplicative*, Trans. Amer. Math. Soc. **364** (2012), no. 2, 933–959, doi:[10.1090/S0002-9947-2011-05479-6](https://doi.org/10.1090/S0002-9947-2011-05479-6); arXiv:[1003.2221v2](https://arxiv.org/abs/1003.2221v2). Theorems 1.5–1.6 are cited with the arXiv v2 pagination, p. 2.
- [16] B. Adamczewski, J. Bell and D. Smertnig, *A height gap theorem for coefficients of Mahler functions*, J. Eur. Math. Soc. **25** (2023), no. 7, 2525–2571, doi:[10.4171/JEMS/1244](https://doi.org/10.4171/JEMS/1244). Theorem 1.2(b), p. 2528; logarithmic height in Section 1.1.1, p. 2527.